MILITARY SPECIFICATION
INITIATOR, CARTRIDGE ACTUATED, DELAY, M10, ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Cartridge Actuated, Delay, M10, Assembly, loaded with Cartridge, Initiator, Delay, M70 for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATION
Military
MIL-A-2550 - Ammunition and Special Weapons, General Specification For

STANDARDS
Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1167 - Ammunition Data Card
MS-28741 - Hose Assembly-Detachable End Fitting, Medium Pressure

DRAWINGS
Munitions Command
F8594861 - Initiator, Cartridge Actuated, Delay, M10, Assembly
C8798142 - Box, Packing, Ammunition for Initiator Delay Cartridge Actuated M10, M30, M31, M32, M42, M43 or M72 with Cartridge in Fiber Container, M398

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2.1 (cont)

DRAWINGS 'Munitions Command'

C8798143 - Container, Ammunition, Fiber, M398 for Initiator Delay Cartridge Actuated M10, M30, M31, M32, M42, M43 or M72 with Cartridge

IEL-8594861 - Inspection Equipment List for Initiator, Cartridge Actuated, Delay, M10, Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Breakaway torque.- The chamber and cap of the assembled initiator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation.

3.3 Residual magnetism.- The initiator shall not deflect the indicator of an approved compass more than three degrees in either direction when passed in front of the compass at a distance of six inches.

3.4 Functional.

3.4.1 Ballistics.- Initiators loaded with accepted cartridges and temperature conditioned shall produce pressure values specified in Table I. Pressure values shall be recorded at the end of 15 feet of aircraft hose, size no. 4, in accordance with Standard MS-28741.

Table I

Temperature	Pressure (psi)	Ignition Delay (see 6.2)
-65° to -70° F	500 to 1250	2.50 seconds, max
65° to 75° F	600 to 1500	1.80 to 2.20 seconds
160° to 165° F	700 to 1700	1.60 seconds, min

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3.4.2 The initiator shall not fail to fire.

3.4.3 The initiator shall function without mechanical failure (see 6.1).

3.5 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects.- The part shall be free from cracks, splits, cold shuts, inclusions, porosity or similar defects.

3.5.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of cartridges.

4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plan and associated criteria in Table II. All nonconforming material shall be rejected. The total number of allowable defects per

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4.2.2 (cont)

class for either the major or minor classification shall not exceed the acceptance number of Table II. Four power magnification may be used for inspection of such defects as burr and foreign matter.

Table II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% examination	-	-	-	-
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3

4.2.2.1 Cartridge, Initiator, Delay, M70, Drawing D8593274.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Case split or perforated	Visual
3. O-ring missing	Visual
4. Sealing disc missing or damaged	Visual
5. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Cartridge discolored, dirty, oily, corroded or stained	Visual
102. Case dented	Visual
103. O-ring split, cut or cracked	Visual
MINOR	
201. Case scratched, wrinkled or bulged	Visual
202. Varnish missing or inadequate	Visual

4.2.2.2 Assembly, Drawing F8594861, less Drawings B8359536, B8593298, D8594856, D8594857 and D8594860.

CRITICAL	
1. Set screw missing	Visual
MAJOR	
101. Thread damaged	Visual
102. Burr	Visual
103. Foreign matter	Visual

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4.2.2.2 (cont)

Defect	Method of Inspection
MINOR	
None defined.	

4.2.2.3 Assembly, F8594861, less B8359536, B8593298 and D8594857.

CRITICAL		
1.	Cartridge loose	Visual/Manual
2.	Cartridge retainer loose	Visual/Manual
3.	Metering groove of cartridge retainer missing	Visual
4.	Elongated hole of cartridge retainer missing	Visual
MAJOR		
None defined.		
MINOR		
None defined.		

4.2.2.4 Assembly, F8594861, less B8359536 and B8593298.

CRITICAL		
1.	Location of cartridge retainer	Gage
2.	Shear pin not engaged in firing pin	Gage
3.	Any part missing	Visual
4.	Improper assembly	Visual
5.	Foreign matter in gas port	Visual
MAJOR		
101.	Length from firing pin housing boss face to gas port face, max	Gage
MINOR		
201.	Varnish missing or inadequate	Visual
202.	Location of label, incorrect	Visual
203.	Barrier material missing from set screw	Visual

4.2.2.5 Assembly, F8594861.

CRITICAL		
1.	Marking missing, incorrect or illegible	Visual
MAJOR		
101.	Height of cap, max	Gage
102.	Cap set screw missing, loose or above flush	Visual

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4.2.2.5 (cont)

	Defect	Method of Inspection
MINOR		
201.	Total length, max	Measure
202.	Location of marking, incorrect	Visual
203.	Shipping cap or plug missing	Visual
204.	Protective finish inadequate or defective	Visual

4.2.2.6 Container, Fiber, Drawing C8798143.

CRITICAL
None defined.

MAJOR
None defined.

MINOR		
201.	Gap between body and cover, max	Manual
202.	Contents loose	Manual
203.	Metal ends loose or distorted	Manual
204.	Cut or damaged through all impregnated layers	Visual
205.	Marking missing, incorrect or illegible	Visual
206.	Tape incomplete or badly wrinkled	Visual
207.	Length of tear tab, min	Visual
208.	Color of tape incorrect	Visual

4.2.2.7 Box, Packing, Drawing C8798142.

CRITICAL
None defined.

MAJOR
None defined.

MINOR		
201.	Contents loose	Manual
202.	Hardware missing, loose, broken or improperly engaged	Visual/Manual
203.	Strapping missing, loose, broken or improperly assembled	Visual/Manual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Marking missing, incorrect or illegible	Visual

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Defect	Method of Inspection
MINOR	
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque.- The initiator chamber and cap of each assembled initiator shall be subjected to the breakaway torque application. Failure of any assembly to comply with the requirements of 3.2 shall be cause for rejection of the assembly.

4.2.3.2 Residual magnetism.- Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any assembly to comply with the requirements of 3.3 shall be cause for rejection of the assembly.

4.2.3.3 Functional.- The following tests shall be performed by the Government:

4.2.3.3.1 Ballistics.- Initiators shall be selected for ballistic tests in the quantity specified in Table III. Failure of any initiator to comply with the requirements of 3.4.1, 3.4.2 or 3.4.3 shall be cause for rejection of the lot of initiators.

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-8658384.

4.3 Test methods and procedures.

4.3.1 Breakaway torque.- A torque wrench, set at the torque requirements specified in 3.2, shall be used in conjunction with an approved fixture to determine the adequacy of the joint. The initiator chamber shall be secured in an air vise or other approved device. The torque shall be applied to the cap.

4.3.2 Residual magnetism.- The initiator shall be tested for excessive residual magnetism in an area free of local magnetic effects, by placing the initiator six inches from and at the same height as an approved compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading as shown in figure 1. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90-degree rotation of the initiator.

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4.3.3 Functional.

4.3.3.1 Ballistics.- Samples shall be selected, divided into test groups and conditioned in accordance with the provisions of Table III.

Table III

First Production Lot

		Conditioning Temperature		
Lot Size	No. of Samples	-65° to -70° F	65° to 75° F	160° to 165° F
1 - 800	15	5	5	5
801 - 1800	30	10	10	10
		All Subsequent Lots		
1 - 800	10	6	2	2
801 - 1800	20	12	4	4

Upon completion of the required temperature conditioning, the samples shall be ballistically tested within five minutes after removal from the conditioning chamber (see Figure 2). Samples shall be reconditioned if this time is exceeded. When reconditioning initiators at minus 65 degrees F, all condensation shall be removed from the interior surfaces before returning the initiator to the conditioning chamber.

4.3.3.2 Temperature conditioning.- A standard M10 Initiator, loaded with a dummy M70 Cartridge, containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1-hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

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5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level A.- Packing and marking shall be in accordance with Drawings C8798142 and C8798143.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 Ignition delay is defined as the time that elapses between the firing of the percussion element and the beginning of continuous pressure rise on the pressure time record.

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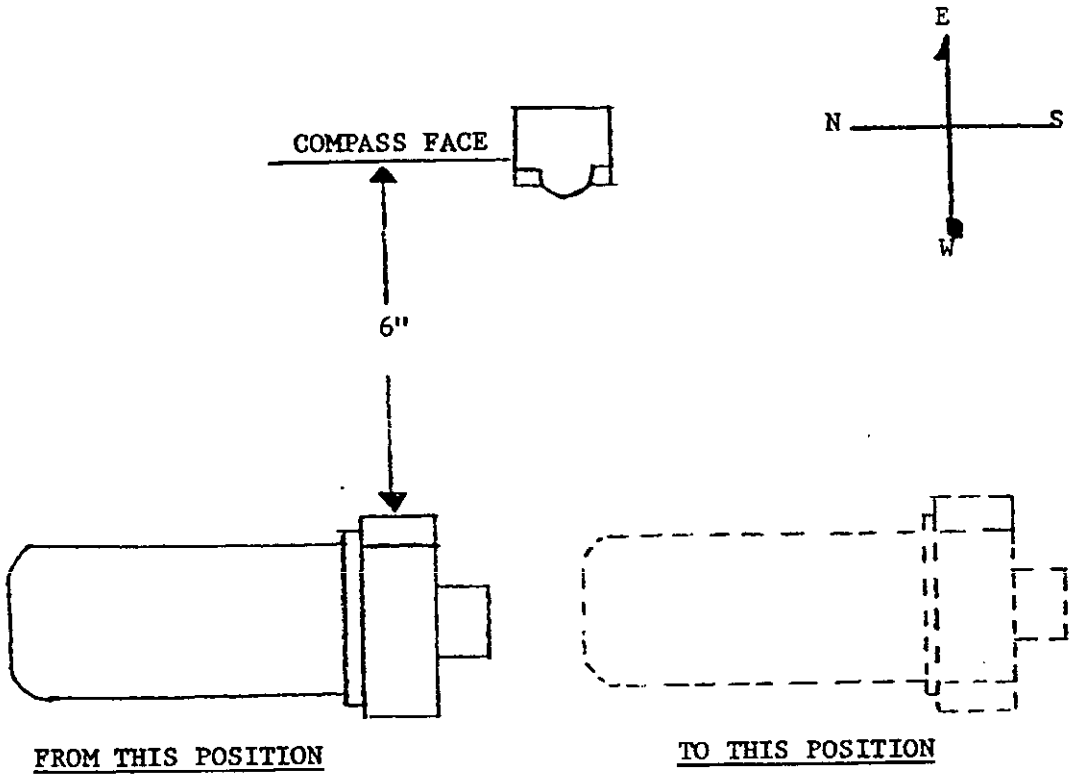
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PILOT'S STANDBY COMPASS, TYPE AN-5766



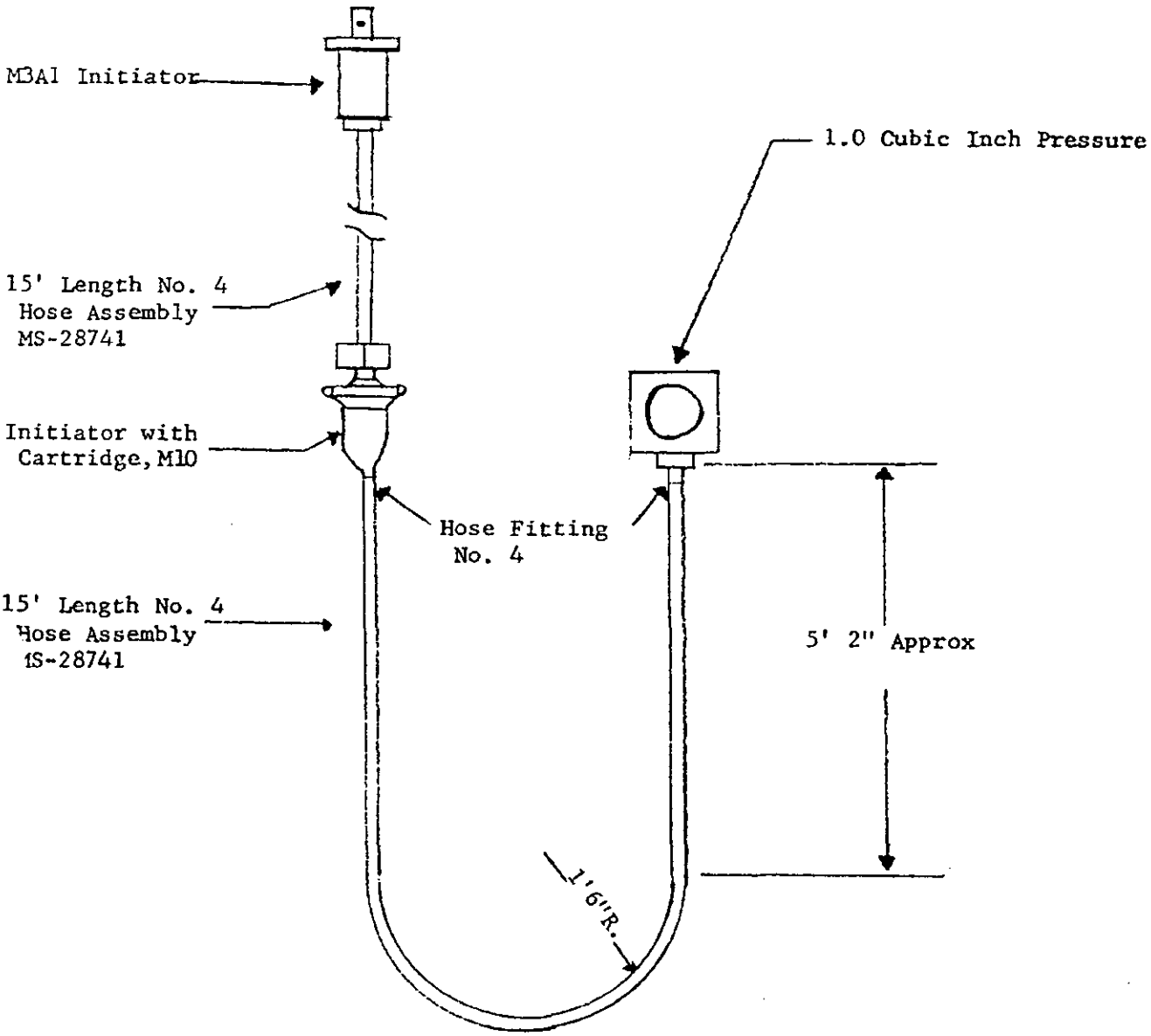
PLAN VIEW

NOTE: INITIATOR AND COMPASS SHALL LIE IN A HORIZONTAL PLANE

LAYOUT FOR DETERMINATION OF RESIDUAL MAGNETISM IN INITIATOR, DELAY, M10

Figure 1

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PERFORMANCE TEST LAYOUT SKETCH

TOP VIEW

Figure 2

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